

The Structural Polynomial of a Parity Sequence and Its Applications

Jon Seymour
jon@wildducktheories.com
Sydney, Australia

July 2026

Abstract

We develop a polynomial representation for parity sequences of generalised Collatz-type maps (g, h, q) , working in the ring $\mathbb{Z}[g, h]$ rather than specialising immediately to numerical values. The central object is the *structural polynomial* $k_w(g, h)$ attached to a parity sequence, which encodes the accumulated additive contribution of the odd steps independently of the parameter q .

The main results are: (1) a *path identity* $h^e b = g^o a + q k_w(g, h)$ valid for any parity sequence; (2) a *geometric polynomial theorem* showing that Steiner words $\text{St}(\alpha, \beta)$ have $k(g, h) = (g^\alpha - h^\alpha)/(g - h)$, independent of β ; (3) a *composition rule* $k_{Q \circ P} = g^{o_Q} k_P + h^{e_P} k_Q$; and (4) a *cycle element identity* $(h^e - g^o) \cdot a = q \cdot k_w(g, h)$ giving the fixed point in terms of (g, h, q) .

Beyond these core results, the path identity is read as a congruence constraint, yielding an *affine path family* parameterised by an *admissibility residue* r ; cycles arise precisely when this family intersects the diagonal $a = b$. Integer solutions to the cycle element identity are classified into four *cycle flavours* (natural, reduced, negative, rational), each an independently defined normalisation of the solution pair (a, q) . We also develop a system of five equivalent representations of any parity sequence — its bits (the p -value), the structural polynomial $k_w(g, h)$, its specialisation $k_w(3, 2)$, the σ -polynomial $\sigma_w(u, v)$, and the orbit parity sequence ϕ_w — connected by explicitly invertible transformations. We introduce a successor operation on k_w in $\mathbb{Z}/(h^e - g^o)\mathbb{Z}$ that generates all cycle representatives by rotation, and we warn that polynomial non-divisibility of $h^e - g^o \nmid k_w$ in $\mathbb{Z}[g, h]$ does not preclude pointwise divisibility at specific integer bases. A concluding section reflects on what the abstract path framework reveals and what it cannot settle. Applications to the $2a + 1$ composition identity are developed in the companion paper [4].

Contents

1	Introduction	2
2	The Structural Polynomial and Path Identity	3

3	Steiner Words and the Geometric Polynomial Theorem	5
4	Composition of Structural Polynomials	5
5	Affine Path Equations	6
5.1	The admissibility constraint	6
5.2	The affine path family	7
5.3	Intersection with cycles	7
6	The Cycle Element Identity	8
7	Cycle Flavours	13
8	Representations of a Parity Sequence	14
8.1	The p -value (edges B, C; centre node)	16
8.2	The σ -polynomial (edges A, D)	17
8.3	Worked example (all edges, $w = OEEOEOEE$)	18
9	Examples	19
10	What the Abstract Path Tells Us — and What It Does Not	20
11	Related Work	22

1 Introduction

A *generalised Collatz-type map* with basis (g, h, q) consists of two elementary operations on the positive integers: an *odd step* $x \mapsto gx + q$ and an *even step* $x \mapsto x/h$. The standard Collatz map (the Syracuse odd-only iteration) uses basis $(g, h, q) = (3, 2, 1)$; the $5x + 1$ analogue uses $(5, 2, 1)$. A *parity sequence* is a finite word over $\{O, E\}$ specifying the order in which these steps are applied. Each parity sequence acts as an affine map on its domain of definition.

The central algebraic object we study is the *structural polynomial* $k(g, h) \in \mathbb{Z}[g, h]$ attached to a parity sequence. It captures the additive part of the parity sequence's affine action in a basis-independent form. The key advantage of working in $\mathbb{Z}[g, h]$ rather than substituting $g = 3, h = 2$ at the outset is that polynomial identities can be proved and, just as importantly, their *failure for other bases* can be identified precisely. A numerical check that an identity holds at $(3, 2)$ gives no information about why it fails at $(5, 2)$; working symbolically makes the obstruction explicit.

The power of the symbolic approach is illustrated dramatically by the $5x + 1$ system, which possesses a non-trivial cycle $17 \rightarrow 43 \rightarrow 27 \rightarrow 17$ under the Syracuse $5x + 1$ map. Via the cycle element identity (Section 6), the fixed point $a = 17$ is determined by $(h^e - g^o) \cdot a = q \cdot k(g, h)$: with $k(5, 2) = 51$ and $h^7 - g^3 = 128 - 125 = 3$, we get $3 \cdot a = 51$, giving $a = 17$. The near-miss $h^e - g^o = 3$ is the arithmetic miracle that makes the cycle possible, and it is visible as a polynomial phenomenon.

The paper is organised as follows. Section 2 establishes the path identity and defines the structural polynomial. Section 3 proves the geometric polynomial theorem for Steiner words. Section 4 derives the composition rule. Section 5 develops the affine path family, the admissibility residue, and their intersection with cycles. Section 6 develops the cycle element identity as an algebraic object: natural and reduced encodings, the uniform factor lemma, the successor operation, and a proposition cautioning that polynomial non-divisibility does not imply pointwise non-divisibility. Section 7 classifies cycle encodings into four flavours (natural, reduced, negative, rational). Section 8 develops the five representations of a parity sequence — the orbit parity sequence ϕ_w , the p -value, the σ -polynomial, $k_w(g, h)$, and $k_w(3, 2)$ — and unifies them in a single diagram. Section 9 collects worked examples illustrating the theory. Section 10 reflects on what the abstract path framework reveals and what it leaves open.

2 The Structural Polynomial and Path Identity

Definition 2.1 (Basis and elementary maps). A *basis* is a triple (g, h, q) of integers with $g, h \geq 1$ and $q \neq 0$. The associated elementary maps are the *odd step* $O: x \mapsto gx + q$ and the *even step* $E: x \mapsto x/h$.

Definition 2.2 (Parity sequence). A *parity sequence* is a finite word $w = w_1 w_2 \cdots w_n$ (possibly empty, $n \geq 0$) over the alphabet $\{O, E\}$. It defines an affine map Φ_w by composing the corresponding elementary maps from left to right: $\Phi_w = \Phi_{w_n} \circ \cdots \circ \Phi_{w_1}$ (the empty composition is the identity map). We write $o = o(w)$ for the number of occurrences of O in w and $e = e(w)$ for the number of occurrences of E .

Remark 2.3. Throughout this paper we assume $h \mid x$ whenever an E -step is applied, so that all maps are defined on their natural domain.

Definition 2.4 (κ -tuple). Let w be a parity sequence with $o \geq 1$ odd steps and e even steps. Number the O -steps from 0 to $o - 1$ in the order they appear in w . For $0 \leq j \leq o - 1$, let κ_j denote the number of E -steps that appear in w *strictly before* the $(j + 1)$ -th O -step. The sequence $(\kappa_0, \kappa_1, \dots, \kappa_{o-1})$ is the κ -tuple of w . It satisfies $0 \leq \kappa_0 \leq \kappa_1 \leq \cdots \leq \kappa_{o-1} \leq e$. For example, $w = OEEOEOEE$ has $o = 3$ odd steps at positions 0, 3, 5, preceded by 0, 2, 3 even steps respectively, so $\kappa = (0, 2, 3)$.

Remark 2.5. For the standard Collatz basis ($g = 3, q = 1$), a parity sequence beginning with an O -step (i.e. $\kappa_0 = 0$) corresponds to a *legal* orbit (one driven by the actual parity of each element, not by the letters of w) starting from an odd integer if and only if its κ -tuple is strictly increasing (equivalently, no two O -steps are adjacent). The polynomial framework works unchanged for non-strict profiles (see Example 9.3).

Definition 2.6 (Structural polynomial). The *structural polynomial* of a parity sequence w with κ -tuple $(\kappa_0, \dots, \kappa_{o-1})$ is

$$k_w(g, h) = \sum_{j=0}^{o-1} g^{o-1-j} h^{\kappa_j} \in \mathbb{Z}[g, h].$$

When $o = 0$, we set $k_w = 0$.

We can now state and prove the central algebraic identity.

Theorem 2.7 (Path identity). *Let (g, h, q) be a basis and w a parity sequence with o odd steps and e even steps, having structural polynomial $k_w(g, h)$. If $b = \Phi_w(a)$ then*

$$h^e \cdot b = g^o \cdot a + q \cdot k_w(g, h). \quad (1)$$

Proof. We proceed by induction on the length $n = |w|$ of the parity sequence.

Base case ($n = 0$). The empty word has $o = e = 0$, $k_\varepsilon = 0$, and $\Phi_\varepsilon = \text{id}$, so $b = a$ and the identity reads $1 \cdot a = 1 \cdot a + q \cdot 0$, which holds.

Inductive step for single-letter words. If $w = O$, then $e = 0$, $o = 1$, $b = ga + q$, and $k_w = 1$. The identity reads $b = ga + q$, which holds. If $w = E$, then $o = 0$, $e = 1$, $b = a/h$, $k_w = 0$, and the identity reads $h \cdot (a/h) = a$, which holds.

Inductive step. Suppose (1) holds for all parity sequences of length less than n , and write $w = w' \cdot \sigma$ where $\sigma \in \{O, E\}$ is the last step. Let $c = \Phi_{w'}(a)$ be the intermediate value, so $b = \Phi_\sigma(c)$. By the inductive hypothesis,

$$h^{e'} \cdot c = g^{o'} \cdot a + q \cdot k_{w'}(g, h), \quad (2)$$

where $o' = o(w')$ and $e' = e(w')$.

Case $\sigma = E$. Then $b = c/h$, $o = o'$, and $e = e' + 1$. Since $h^{e'+1}(c/h) = h^{e'}c = g^{o'}a + qk_{w'}$ by (2), and the E-step appended at the end does not precede any O-step, so the κ -tuple of w equals that of w' , and therefore $k_w = k_{w'}$. The identity (1) holds.

Case $\sigma = O$. Then $b = gc + q$, so $o = o' + 1$ and $e = e'$. The new O-step is the last (index $j = o'$) in w . All $e' = e$ even steps in w precede it (they all belong to w'), so $\kappa_{o'} = e'$. The structural polynomial of w is

$$k_w(g, h) = g \cdot k_{w'}(g, h) + h^{e'},$$

because the exponents of the first o' terms each gain one extra factor of g , and the new final O-step contributes $g^0 h^{e'} = h^{e'}$. Now compute:

$$\begin{aligned} h^e \cdot b &= h^{e'} \cdot (gc + q) \\ &= g \cdot h^{e'} c + q \cdot h^{e'} \\ &= g(g^{o'} a + q \cdot k_{w'}(g, h)) + q \cdot h^{e'} \quad (\text{by (2)}) \\ &= g^{o'+1} a + q(g \cdot k_{w'}(g, h) + h^{e'}) \\ &= g^o \cdot a + q \cdot k_w(g, h). \end{aligned} \quad \square$$

Remark 2.8. Two features of the path identity deserve emphasis.

1. The polynomial $k_w(g, h)$ is independent of q : the parameter q appears only as a uniform scalar multiplier. Thus the additive structure of parity sequences is entirely captured by the polynomial ring $\mathbb{Z}[g, h]$.
2. The polynomial $k_w(g, h)$ is independent of trailing E -steps: appending E -steps at the end only scales h^e on the left-hand side of (1), leaving k_w unchanged. This means k_w alone does not uniquely identify a parity sequence; the p -value (Section 8) resolves the ambiguity by appending a sentinel bit that records the length n .

3 Steiner Words and the Geometric Polynomial Theorem

Definition 3.1 (Steiner word). For integers $\alpha \geq 1$ and $\beta \geq 0$, the *Steiner word* $\text{St}(\alpha, \beta)$ is the parity sequence

$$\text{St}(\alpha, \beta) = \underbrace{(OE)^\alpha}_{\alpha \text{ copies of } OE} \underbrace{E^\beta}_{\beta \text{ extra } E\text{-steps}}.$$

It has $o = \alpha$ odd steps and $e = \alpha + \beta$ even steps.

Remark 3.2. Steiner words are the natural building blocks of Collatz orbits under the Syracuse (odd-only) map: each Syracuse step $S(a) = (3a + 1)/2^{v_2(3a+1)}$ corresponds to a Steiner word $\text{St}(1, \beta)$ where $\beta = v_2(3a + 1) - 1 \geq 0$. A block of α consecutive applications of S before the next element divisible by an extra power of h corresponds to $\text{St}(\alpha, \beta)$.

The κ -tuple of $\text{St}(\alpha, \beta)$ has a particularly simple form.

Lemma 3.3 (κ -tuple of a Steiner word). *The κ -tuple of $\text{St}(\alpha, \beta)$ is $\kappa_j = j$ for $0 \leq j \leq \alpha - 1$.*

Proof. In $(OE)^\alpha E^\beta$, the j -th O -step (zero-indexed) appears at position $2j + 1$. Exactly j E -steps appear before it (one after each of the first j O -steps). Hence $\kappa_j = j$. $\square$

Theorem 3.4 (Geometric polynomial theorem). *For any $\alpha \geq 1$ and $\beta \geq 0$,*

$$k_{\text{St}(\alpha, \beta)}(g, h) = \frac{g^\alpha - h^\alpha}{g - h} = \sum_{j=0}^{\alpha-1} g^{\alpha-1-j} h^j. \quad (3)$$

In particular, $k_{\text{St}(\alpha, \beta)}$ is independent of β .

Proof. Substituting $\kappa_j = j$ from Lemma 3.3 into Definition 2.6:

$$k_{\text{St}(\alpha, \beta)}(g, h) = \sum_{j=0}^{\alpha-1} g^{\alpha-1-j} h^j = \frac{g^\alpha - h^\alpha}{g - h},$$

where the last equality is the standard geometric series identity. Independence of β follows from Remark 2.8. $\square$

4 Composition of Structural Polynomials

If P and Q are parity sequences, the composition $Q \circ P$ denotes the parity sequence obtained by concatenating P then Q : first apply P , then apply Q to the result.

Theorem 4.1 (Composition rule). *Let P and Q be parity sequences with o_P, e_P and o_Q, e_Q odd and even steps respectively. Then*

$$k_{Q \circ P}(g, h) = g^{o_Q} \cdot k_P(g, h) + h^{e_P} \cdot k_Q(g, h). \quad (4)$$

Proof. Let $(\kappa_0^P, \dots, \kappa_{o_P-1}^P)$ and $(\kappa_0^Q, \dots, \kappa_{o_Q-1}^Q)$ be the κ -tuples of P and Q respectively.

In the concatenated sequence $Q \circ P$, the first o_P odd steps come from P and the last o_Q odd steps come from Q . The E -steps of P all precede the E -steps of Q .

For the j -th odd step of P ($0 \leq j < o_P$): the number of E -steps preceding it in $Q \circ P$ equals κ_j^P (no E -steps from Q are involved).

For the j -th odd step of Q ($0 \leq j < o_Q$): the number of E -steps preceding it in $Q \circ P$ is $e_P + \kappa_j^Q$ (all e_P steps of P plus the κ_j^Q steps of Q that precede it).

Therefore the structural polynomial of $Q \circ P$ is

$$\begin{aligned} k_{Q \circ P}(g, h) &= \sum_{j=0}^{o_P-1} g^{(o_P+o_Q-1)-j} h^{\kappa_j^P} + \sum_{j=0}^{o_Q-1} g^{o_Q-1-j} h^{e_P+\kappa_j^Q} \\ &= g^{o_Q} \sum_{j=0}^{o_P-1} g^{o_P-1-j} h^{\kappa_j^P} + h^{e_P} \sum_{j=0}^{o_Q-1} g^{o_Q-1-j} h^{\kappa_j^Q} \\ &= g^{o_Q} \cdot k_P(g, h) + h^{e_P} \cdot k_Q(g, h). \end{aligned} \quad \square$$

Remark 4.2 (Interpretation). The two terms in (4) have natural interpretations. The factor g^{o_Q} on k_P accounts for the fact that after P has run, Q applies o_Q further odd multiplications, each of which scales the additive accumulation from P by g . The factor h^{e_P} on k_Q accounts for the fact that Q 's own E -accumulation exponents are displaced upward by the e_P even divisions already consumed by P .

The composition rule tells us how structural polynomials combine, but it leaves open a complementary question: given a fixed word w , which starting values a are compatible with it as a path, and which make it a cycle? Section 5 answers this by reading the path identity as a congruence constraint on a .

5 Affine Path Equations

The path identity (1) can be read in two directions. Read as a statement about a specific pair (a, b) , it asserts that starting from a and following word w arrives at b . Read as a constraint, it characterises the full family of integer pairs (a, b) for which w is a valid path.

5.1 The admissibility constraint

Throughout this section we assume $\gcd(g, h) = 1$.

For a word w with o odd steps and e even steps, the path identity is

$$h^e \cdot b = g^o \cdot a + q \cdot k_w(g, h).$$

For b to be a positive integer we need $h^e \mid g^o a + q k_w(g, h)$. Since $\gcd(g^o, h^e) = 1$ when $\gcd(g, h) = 1$, this is a congruence condition on a alone:

$$g^o \cdot a \equiv -q k_w(g, h) \pmod{h^e}.$$

Because $\gcd(g, h) = 1$ implies g^o is invertible modulo h^e , this has a unique solution $r \in \{0, 1, \dots, h^e - 1\}$:

$$r \equiv -qk_w(g, h) \cdot (g^o)^{-1} \pmod{h^e}.$$

We call r the *admissibility residue* of w . (This use of “admissibility” is distinct from the orbit-level sense in the remark after Definition 2.4: here it means integrality of each E-step’s output, not legality of the driving parity sequence.)

5.2 The affine path family

Every valid starting point a for a path with word w satisfies $a \equiv r \pmod{h^e}$, so the family of integer pairs is parameterised by $t \geq 0$:

$$a(t) = h^e \cdot t + r, \quad b(t) = g^o \cdot t + b_0, \quad (5)$$

where $b_0 = (g^o r + qk_w(g, h))/h^e$ is the image of r under Φ_w . Since $r \in \{0, 1, \dots, h^e - 1\}$ (the canonical representative chosen above), $a(t) = h^e t + r \geq 0$ for all $t \geq 0$, with $a(t) > 0$ whenever $t \geq 1$ or $r > 0$. Both $a(t)$ and $b(t)$ are affine functions of t ; the path $a \rightarrow b$ is the letter-driven trajectory prescribed by w .

Remark 5.1. The residue r encodes the admissibility of the path: a starting value a is compatible with word w if and only if $a \equiv r \pmod{h^e}$. This is the modular shadow of the requirement that each prescribed E-step is applied to an even value.

Example 5.2 (St(2, 1) at the Collatz basis). For $w = \text{St}(2, 1) = \text{OEEOE}$ at $(g, h, q) = (3, 2, 1)$: $o = 2$, $e = 3$, $k_w(3, 2) = g + h = 5$, $h^e = 8$, $g^o = 9$. Since $9 \equiv 1 \pmod{8}$, the admissibility residue is $r \equiv -5 \cdot 1 \equiv 3 \pmod{8}$. The affine family is $a(t) = 8t + 3$, $b(t) = 9t + 4$, giving the paths $3 \rightarrow 4$, $11 \rightarrow 13$, $19 \rightarrow 22$, $27 \rightarrow 31$, $\dots$

5.3 Intersection with cycles

Setting $b = a$ in (5) requires

$$h^e \cdot t + r = g^o \cdot t + b_0,$$

hence $(h^e - g^o) \cdot t = b_0 - r$, i.e. $d \cdot t = b_0 - r$. This has an integer solution in t if and only if $d \mid (b_0 - r)$, and when it does, the solution is unique; call it $t_0 = (b_0 - r)/d$, giving the cycle element $a_0 = h^e t_0 + r$. Substituting and using $b_0 = (g^o r + qk)/h^e$:

$$d \cdot a_0 = h^e b_0 - g^o r = (g^o r + qk) - g^o r = qk.$$

The admissibility residue r cancels exactly. We record this as a corollary.

Corollary 5.3 (Cycle element identity). *Let w be a parity sequence with o odd steps, e even steps, and $d = h^e - g^o \neq 0$. If the affine path family (5) intersects the diagonal $a = b$ at a non-negative integer t_0 , then the cycle element $a_0 = h^e t_0 + r$ satisfies*

$$d \cdot a_0 = q \cdot k_w(g, h). \quad (6)$$

The defining congruence for r gives $h^e \mid g^o r + qk$, so $b_0 = (g^o r + qk)/h^e$ is an integer. The admissibility residue r does not appear in (6): if $d \mid b_0 - r$, the integer solution $t_0 = (b_0 - r)/d$

is unique because $d \neq 0$. Hence a_0 is the unique fixed point of Φ_w associated with this word w and this admissibility residue r . Other elements of a nontrivial orbit correspond to rotations of w ; the fixed-point value a_0 is determined entirely by w and the basis (g, h, q) via (6).

The affine path family thus makes explicit what the cycle identity summarises: a cycle exists when the path family and the diagonal $a = b$ intersect at a non-negative integer t_0 , and at that intersection the admissibility residue drops out. Section 6 develops this identity as an algebraic object in its own right.

Remark 5.4. For path problems, t ranges over $\mathbb{Z}_{\geq 0}$: since $r \in \{0, 1, \dots, h^e - 1\}$, every $t \geq 0$ gives $a(t) \geq 0$, and the full family is an infinite arithmetic progression of valid (a, b) pairs. For cycle problems, t is pinned to a single determined value t_0 (if an integer solution exists). The transition from path to cycle is therefore a transition from a free parameter to a determined one — the same phenomenon seen in the cycle element identity, here made geometric.

Example 5.5 (The $5x + 1$ cycle at $a = 17$). Consider the clause $w = \text{OEEOEEEOEEE}$ at basis $(g, h, q) = (5, 2, 1)$: $o = 3$, $e = 7$, $h^e = 128$, $g^o = 125$, $d = 3$. The κ -tuple is $(0, 1, 4)$, giving $k_w(g, h) = g^2 + gh + h^4$ and $k_w(5, 2) = 25 + 10 + 16 = 51$.

The admissibility residue is $r \equiv -51 \cdot (125)^{-1} \pmod{128}$. Since $125 \equiv -3 \pmod{128}$ and $(-3)^{-1} \equiv 85 \pmod{128}$ (as $(-3)(85) = -255 \equiv 1 \pmod{128}$):

$$r \equiv -51 \cdot 85 = -4335 \equiv 17 \pmod{128}.$$

The image of r under Φ_w is $b_0 = (125 \cdot 17 + 51)/128 = 17$, so the affine family is

$$a(t) = 128t + 17, \quad b(t) = 125t + 17.$$

Setting $a = b$: $3t = b_0 - r = 0$, giving $t_0 = 0$ and cycle element $a_0 = 17$. The equation $3t = 0$ has exactly one non-negative integer solution, so 17 is the unique element of this cycle.

6 The Cycle Element Identity

Section 5 derived the cycle element identity as a geometric consequence: setting $a = b$ in the affine path family forces $d \cdot a = qk$. We now develop this identity as an algebraic object in its own right, establishing the reduction theory and the successor operation that generate all cycle representatives.

Corollary 5.3 established the *cycle element identity* geometrically: any a with $\Phi_w(a) = a$ satisfies

$$(h^e - g^o) \cdot a = q \cdot k_w(g, h). \tag{7}$$

Writing $d = h^e - g^o$ and $k = k_w(g, h)$, the *natural solution* $(a, q) = (k, d)$ is always an integer solution, so every abstract cycle word is immediately realisable. The interesting question is which cycles admit a *reduced* solution — one where $\gcd(a, q) = 1$ and in particular $q = 1$. We develop the reduction theory in four stages: (1) the natural encoding (Proposition 6.1); (2) the definition of the joint $\gcd$ δ_w (Definition 6.3) and the uniform factor lemma for coprime bases (Lemma 6.4); (3) the behaviour of δ_w for non-coprime bases (Lemma 6.6); and (4) the reduced identity and its consequences (Theorems 6.8–6.10, Propositions 6.11–6.12).

Throughout this section $\gcd$ denotes the positive greatest common divisor. All quantities $k_i = k_{w_i}(g, h)$ and $d = h^e - g^o$ are integers evaluated at a fixed basis (g, h) ; the $\gcd$ is taken over these integer values.

Proposition 6.1 (Every abstract cycle has a natural encoding). *Let w be any parity sequence with $o \geq 1$ odd steps and e even steps, and let (g, h) be a pair of positive integers with $d = h^e - g^o \neq 0$. Setting $q = d$ and $a = k = k_w(g, h)$, the pair (a, q) satisfies $\Phi_w(a) = a$ at basis (g, h, q) . In particular, every abstract cycle word w is realised as a genuine integer cycle at every pair (g, h) with $h^e \neq g^o$, simply by choosing $q = h^e - g^o$.*

Proof. The pair $(a, q) = (k, d)$ satisfies $d \cdot k = d \cdot k$ trivially, hence $d \cdot a = q \cdot k$. By the path identity (1), $h^e \cdot \Phi_w(k) = g^o \cdot k + d \cdot k = (g^o + d)k = h^e k$, and since $h > 0$ we may cancel to obtain $\Phi_w(k) = k$.

It remains to verify that every intermediate E-step is admissible, i.e. that the running value is always divisible by h when an E-step is applied. The intermediate states are precisely the rotation structural values $k_0 = k, k_1, \dots, k_{n-1}$. For an O-leading rotation, $k_{i+1} = gk_i + d$ (Proposition 6.12), which is an integer. For an E-leading rotation, the structural polynomial definition gives $k_{i+1} = k_i/h$, so $h \mid k_i$ and the division is admissible. Thus the orbit $k \rightarrow k_1 \rightarrow \dots \rightarrow k_{n-1} \rightarrow k$ consists entirely of integers and all E-steps are admissible. $\square$

Remark 6.2. Since the natural solution exists for every (g, h) and every word w , the existence of an integer cycle is not in itself remarkable — it is guaranteed by construction once $q = d$. What matters is reduction, developed in Theorem 6.8.

Definition 6.3 (Joint $\gcd$ of a cycle at a basis). Let $w = w_0 w_1 \dots w_{n-1}$ be the complete parity sequence of a cycle — the full closed sentence traversed by a periodic orbit — with o odd steps and e even steps in total, and let (g, h) be a pair of positive integers. The n rotations $w_0, w_1, \dots, w_{n-1}$ are the n indexed re-readings of the same cycle starting from each position in turn (not necessarily distinct if w is a proper power); each rotation w_i is again a complete sentence of length n . Write $k_i = k_{w_i}(g, h) \in \mathbb{Z}$ for the structural value of the i -th rotation evaluated at (g, h) , and $d = h^e - g^o \in \mathbb{Z}$. The *joint $\gcd$ of the cycle w at (g, h)* is

$$\delta_w := \gcd(k_0, k_1, \dots, k_{n-1}, |d|) \in \mathbb{Z}_{>0}.$$

This is a property of the complete cycle, not of any individual rotation or fragment: omitting any rotation can only increase the value of δ_w .

Lemma 6.4 (Coprime bases: single-rotation $\gcd$ equals joint $\gcd$). *Let $\gcd(g, h) = 1$, let w be a parity sequence with $o \geq 1$, and let $k_i = k_{w_i}(g, h)$ and $d = h^e - g^o$. Then $\gcd(k_i, |d|) = \gcd(k_j, |d|)$ for all i, j , and*

$$\delta_w = \gcd(k_i, |d|) \quad \text{for any rotation } i.$$

In particular $\gcd(a_w, |q_w|) = 1$.

Proof. Set $f = \gcd(k_0, |d|)$. We show $f \mid k_i$ for every i by induction on rotation steps.

O-leading step (w_i begins with O): $k_{i+1} = gk_i + d$ (Proposition 6.12). Since $f \mid k_i$ and $f \mid d$, we have $f \mid k_{i+1}$.

E-leading step (w_i begins with E): $k_{i+1} = k_i/h$. Since $\gcd(g, h) = 1$, $\gcd(h, d) = \gcd(h, h^e - g^o) = \gcd(h, g^o) = 1$, so $\gcd(f, h) = 1$. With $f \mid k_i$ and $h \mid k_i$, coprimality gives $f \mid k_i/h = k_{i+1}$.

Hence $f \mid k_i$ for all i , so $f \mid \delta_w$. But $\delta_w \mid k_0$ and $\delta_w \mid |d|$ by definition, hence $\delta_w \mid f$. Therefore $\delta_w = f = \gcd(k_i, |d|)$ for any i , and $\gcd(a_w, |q_w|) = \gcd(k/\delta_w, |d/\delta_w|) = 1$. $\square$

Remark 6.5. This lemma explains the role of $\gcd(g, h) = 1$: the propagation argument fails at E-leading steps when $\gcd(h, d) > 1$, so $f \mid k_i$ need not imply $f \mid k_i/h$. Readers focused on the standard Collatz basis, where $\gcd(3, 2) = 1$ always holds, may skip the next subsection and proceed directly to Theorem 6.8.

The non-coprime case

Lemma 6.6 (Non-coprime bases: per-rotation gcds are inconsistent). *Let $w = w_0 w_1 \cdots w_{n-1}$ be a parity sequence with o odd steps, e even steps, $d = h^e - g^o$, and let $r = \gcd(g, h)$. If $r > 1$ then $r \mid d$, and $\gcd(k_i, |d|)$ varies across rotations in general:*

$$\gcd(k_i, |d|) \neq \gcd(k_j, |d|)$$

for some pairs of rotations $i \neq j$. In particular no single-rotation gcd equals δ_w in general, so the reduced solution requires the full joint gcd δ_w (Definition 6.3).

Proof. Since $r \mid g$ and $r \mid h$, we have $r \mid g^o$ and $r \mid h^e$, hence $r \mid d = h^e - g^o$.

For a monomial $g^{o-1-j} h^{\kappa_j}$ in $k_{w_i}(g, h)$: if $o-1-j > 0$ or $\kappa_j > 0$ then r divides that monomial. The sole exception is $j = o-1$ and $\kappa_{o-1} = 0$, which occurs exactly when rotation w_i has the form $O^o E^e$ (all O-steps precede all E-steps). In that case $k_{w_i}(g, h) = \sum_{j=0}^{o-1} g^{o-1-j} = (g^o - 1)/(g - 1)$, which need not be divisible by r .

Hence $r \mid k_i$ for every rotation that is not of the form $O^o E^e$, but may fail for the block rotation. Since $r \mid d$ regardless, the quantity r divides $\gcd(k_i, |d|)$ for those rotations where $r \mid k_i$ but not necessarily for the block rotation. The per-rotation gcds are therefore inconsistent across rotations, and the joint gcd δ_w — which takes the minimum over all rotations — is the correct divisor. $\square$

Remark 6.7. The failure of the single-rotation formula when $\gcd(g, h) > 1$ is illustrated concretely by $w = \text{OEOEEOE}$ at $(g, h) = (4, 2)$, where $r = 2$. Here $d = 2^5 - 4^3 = -32$, and the eight structural values $k_{w_i}(4, 2)$ are 32, 96, 48, 160, 80, 40, 128, 64.

Per-rotation reduction (using $\gcd(k_i, d)$ for each i separately) gives inconsistent results:

Rotation	k_i	$\gcd(k_i, d)$	$a_i = k_i / \gcd$	$q_i = d / \gcd$
OEOEEOE	32	32	1	-1
EOEEOEEO	96	32	3	-1
OEEOEEOE	48	16	3	-2
EEOEEOEO	160	32	5	-1
EOEEOEEO	80	16	5	-2
OEEOEEOE	40	8	5	-4
EEOEEOEO	128	32	4	-1
EOEEOEEO	64	32	2	-1

The q_i values are inconsistent ($-1, -2, -4$) and the a_i values repeat (the value 5 appears three times, 3 twice), so per-rotation reduction does not yield a valid 8-cycle with distinct elements.

Joint-gcd reduction (using $\delta_w = \gcd(32, 96, 48, 160, 80, 40, 128, 64, |d|) = 8$ for all rotations) gives consistent results:

Rotation	k_i	δ_w	$a_i = k_i/\delta_w$	$q_i = d/\delta_w$
OEOEEOE	32	8	4	-4
EOEEEOE	96	8	12	-4
OEEEOEE	48	8	6	-4
EEEOEOE	160	8	20	-4
EOEEEOE	80	8	10	-4
OEEEOEE	40	8	5	-4
EEEOEOE	128	8	16	-4
EOEEEOE	64	8	8	-4

The q_i values are now uniformly -4 and the eight a_i values 4, 12, 6, 20, 10, 5, 16, 8 are all distinct, giving a valid 8-cycle.

Note that in this example the cycle steps are driven by the parity sequence directly, not by the residue of the current element modulo h . When $\gcd(g, h) > 1$, the parity of a_i modulo h is not well-defined as a basis for deciding the next step, so the letter-driven interpretation (where w prescribes which steps are O -steps and which are E -steps) is the only coherent one.

(For comparison, the same word at $(g, h) = (3, 2)$ has $d = 5$, $k = 23$, $\gcd(k, |d|) = 1$, giving $q_w = 5$, $a_w = 23$ — the standard Collatz cycle $[23, 74, 37, 116, 58, 29, 92, 46]$.)

Theorem 6.8 (Reduced cycle element identity). *Let w be a parity sequence with o odd steps, e even steps, $d = h^e - g^o \neq 0$, $k = k_w(g, h)$, and δ_w the joint gcd of the cycle at (g, h) (Definition 6.3). The reduced solution to (7) is*

$$a_w = \frac{k}{\delta_w}, \quad q_w = \frac{d}{\delta_w},$$

Here $a_w > 0$ and q_w has the sign of d . The pair (a_w, q_w) is the minimal positive- a solution: every other solution with $a > 0$ is a positive-integer multiple of (a_w, q_w) .

When $\gcd(g, h) = 1$, Lemma 6.4 guarantees $\delta_w = \gcd(k, |d|)$, and in particular $\gcd(a_w, |q_w|) = 1$.

Proof. The natural solution (k, d) satisfies (7) by inspection. Since $\delta_w \mid k$ and $\delta_w \mid d$, dividing both sides by δ_w gives the integer solution (a_w, q_w) .

For minimality: call (a, q) with $a > 0$ a *consistent encoding* of the cycle if $d \cdot a_i = q \cdot k_i$ holds for every rotation i , where a_i is the cycle element corresponding to rotation w_i . Any consistent encoding satisfies $d \cdot a = q \cdot k$ for the chosen starting rotation. Writing $q/d = a/k$ in lowest terms as p/r with $\gcd(p, r) = 1$, every $a_i = pk_i/r$ must be an integer, so $r \mid k_i$ for every i . Likewise $r \mid d$ (since $q = pd/r$). Hence $r \mid \gcd(k_i, |d|)$ for all i , so $r \mid \delta_w$. The

largest such r is δ_w , giving the minimal consistent encoding $(a_w, q_w) = (k/\delta_w, d/\delta_w)$; every other consistent encoding with $a > 0$ is a positive-integer multiple of (a_w, q_w) .

When $\gcd(g, h) = 1$, Lemma 6.4 gives $\delta_w = \gcd(k, |d|)$, so $\gcd(a_w, |q_w|) = \gcd(k/\delta_w, |d/\delta_w|) = 1$. $\square$

Remark 6.9. The reduced solution (a_w, q_w) is the minimal positive- a encoding of the cycle w at basis (g, h) . A nontrivial positive Collatz cycle would correspond to a legal, primitive (minimal-period) parity word w at $(g, h, q) = (3, 2, 1)$, distinct from the known OEE cycle up to rotation. In the reduced encoding this requires $d = 2^e - 3^o > 0$ and $d \mid k_w(3, 2)$ (so that $q_w = d/\delta_w = 1$); the sign condition $d > 0$ is essential since $d \mid k$ alone gives only $|q_w| = 1$. No such nontrivial positive cycle is known to exist.

Theorem 6.10 (Cycle determines q). *Let w be a parity sequence with o odd steps, e even steps, $d = h^e - g^o$, $k = k_w(g, h)$, and joint gcd δ_w as in Theorem 6.8. Given w and (g, h) , the additive term q is not a free parameter: the reduced value $q_w = d/\delta_w$ is the unique reduced q for which the integer fixed point $a_w = k/\delta_w > 0$ exists. In particular:*

1. q_w has the sign of $d = h^e - g^o$.
2. $q_w = 1$ if and only if $d > 0$ and $d \mid k_w(g, h)$.
3. Every other solution with $a > 0$ uses $q = mq_w$ for some $m \in \mathbb{Z}_{>0}$, with corresponding cycle element ma_w .

Proof. By Theorem 6.8, every solution (a, q) with $a > 0$ to $d \cdot a = q \cdot k$ is a positive-integer multiple of the reduced solution (a_w, q_w) . Hence q is constrained to $\{mq_w : m \in \mathbb{Z}_{>0}\}$; the three stated properties follow directly from the definition of $q_w = d/\gcd(k, |d|)$. $\square$

Proposition 6.11 (Polynomial vs. pointwise divisibility). *Polynomial non-divisibility of $h^e - g^o$ into $k_w(g, h)$ in $\mathbb{Z}[g, h]$ does not imply pointwise non-divisibility at any particular pair of positive integers (g, h) . Consequently, no argument based solely on polynomial identities can rule out the existence of integer cycles at a specific basis.*

Proof. The $5x + 1$ cycle at $a = 17$ with clause $w = \text{OEOEEEEEEEE}$ is a concrete witness (Example 9.1): the polynomial $h^7 - g^3$ does not divide $k_w(g, h) = g^2 + gh + h^4$ in $\mathbb{Z}[g, h]$, yet at $(g, h) = (5, 2)$ one has $2^7 - 5^3 = 3$ and $k_w(5, 2) = 51 = 3 \times 17$, so $3 \mid 51$. The cycle exists because of the arithmetic coincidence $2^7 - 5^3 = 3$, which is invisible at the polynomial level. Any argument ruling out Collatz cycles must therefore work at the level of integer evaluations. $\square$

Proposition 6.12 (Successor on k_w around a cycle). *Let (g, h, q) be a basis with $\gcd(g, h) = 1$, and let $w = w_0 w_1 \cdots w_{n-1}$ be a cyclic parity sequence with o odd steps and e even steps in total, and let $w' = w_1 \cdots w_{n-1} w_0$ be the rotation by one step. Then, working in $\mathbb{Z}/(h^e - g^o)\mathbb{Z}$,*

$$k_{w'}(g, h) \equiv \begin{cases} g \cdot k_w(g, h) & \text{if } k_w(1, 0) = 1 \text{ (} w \text{ begins with an O-step),} \\ h^{-1} \cdot k_w(g, h) & \text{if } k_w(1, 0) = 0 \text{ (} w \text{ begins with an E-step).} \end{cases}$$

In the O -step case, the unreduced product $g \cdot k_w(g, h)$ promotes the leading monomial $g^{o-1} \mapsto g^o$, and the natural recurrence

$$k_{w'}(g, h) = g \cdot k_w(g, h) + (h^e - g^o)$$

replaces this g^o by h^e — exactly the substitution $g^o \equiv h^e \pmod{h^e - g^o}$ — keeping the degree at $o - 1$ in g . In the E -step case, h^{-1} exists in $\mathbb{Z}/(h^e - g^o)\mathbb{Z}$ because $\gcd(g, h) = 1$ implies $\gcd(h, h^e - g^o) = 1$. The modulus $h^e - g^o$ is the cycle denominator from Theorem 6.8, so this operation is only natural in the context of a fixed cycle (o, e) . As a consequence, if w is aperiodic (not a proper power of a shorter word), the n rotations of w produce at most n distinct values of $k_{w'}(g, h)$ in $\mathbb{Z}/(h^e - g^o)\mathbb{Z}$. For a periodic w of period $p \mid n$, there are at most p distinct values. The σ -successor of Section 8 is the same rotation expressed via the change of variables $g = u/v$, $h = v$.

Remark 6.13. When $d \mid k_w$, the reduced solution has $q_w = 1$, recovering the standard Collatz setting. A worked example ($w = OEEOEOEE$, $q_w = 5$, $a_w = 29$) appears in Section 8.

The reduction theory fixes the reduced solution (a_w, q_w) uniquely once (w, g, h) are given. Section 7 steps back and classifies the full family of normalisations of the solution pair (a, q) .

7 Cycle Flavours

The preceding section established that once (w, g, h) are fixed, the reduced solution (a_w, q_w) is unique. But the equation $d \cdot a = q \cdot k$ admits infinitely many integer solutions related by overall scaling and sign, and different conventions on the signs and integrality of a and q arise naturally in different contexts. We call these *cycle flavours*. They are not distinct cycles: they are the same underlying algebraic object viewed through different choices of normalisation.

Flavour	a	q	$\gcd(a , q)$
Natural	$k > 0$	d (any sign)	≥ 1
Reduced	$k/\delta_w > 0$	d/δ_w (any sign)	1 if $\gcd(g, h) = 1$
Negative	< 0	> 0	1
Rational	k/d (any sign)	1	—

Each flavour is an independently defined normalisation of the solutions to $d \cdot a = q \cdot k$:

- **Natural** $\leftrightarrow$ **scaled**: multiplying both a and q by any $m \in \mathbb{Z}_{>0}$ gives another valid solution. The natural solution (k, d) is the representative with $a = k$.
- **Reduced**: dividing the natural solution by the joint $\gcd \delta_w$ gives $a = k/\delta_w > 0$ and $q = d/\delta_w$, with q inheriting the sign of d . This is the minimal positive- a solution. When $\gcd(g, h) = 1$, $\delta_w = \gcd(k, |d|)$ and $\gcd(|a|, |q|) = 1$. This is the normalisation used throughout the paper.

- **Negative:** an independently defined sign-reassignment normalisation: $a < 0$, $q > 0$, $\gcd(|a|, |q|) = 1$. No factor elimination is performed; the sign of the reduced solution is assigned so that $q > 0$ rather than so that $a > 0$.
- **Rational:** setting $q = 1$ gives $a = k/d \in \mathbb{Q}$. This is an integer exactly when $d \mid k$, i.e. when the reduced cycle has $q_w = 1$. The sign of a matches the sign of d .

The preferred flavour in this paper is *reduced* with $a > 0$. The mythical Collatz cycles are the reduced cycles at $(g, h, q) = (3, 2, 1)$ with $o > 1$: they sit at the intersection of the reduced and rational flavours, being the reduced solutions ($\gcd(a, q) = 1$, $a > 0$, since $\gcd(3, 2) = 1$) for which additionally $q = 1$ and $a \in \mathbb{Z}$.

The flavour classification applies to any fixed specialisation (g, h) . Section 8 steps back and asks a different question: how many ways can a parity sequence be represented, independently of any basis?

8 Representations of a Parity Sequence

The preceding sections computed $k_w(g, h)$ for specific families of words — Steiner words, compositions, cycle words. This section asks a different question: how many ways can a single parity sequence w be represented, and how do those representations relate to one another? The answer is five: bits (p_w) , the structural polynomial $k_w(g, h)$, its integer specialisation $k_w(3, 2)$, the σ -polynomial $\sigma_w(u, v)$, and the orbit parity sequence ϕ_w . The diagram below makes all transformations between them explicit. In particular, the specialisation from $k_w(g, h)$ to $k_w(3, 2)$ (edge E) has no direct inverse: recovering $k_w(g, h)$ from $k_w(3, 2)$ requires going via ϕ_w and the bits (the round-trip $F \rightarrow C \rightarrow B$). Traversing edge F requires $d = h^e - g^o$ as metadata alongside $k_w(3, 2)$ and the basis (g, h) : one applies the $(3, 2, d)$ Collatz map starting from $k = k_w(3, 2)$ and reads parity steps until the orbit returns to k . No knowledge of n is needed — the cycle closes automatically. If (o, e) are known rather than d , they suffice since d is computable from them; but $n = o + e$ alone does not determine d and is not what F requires. This indirectness reflects the distinction between polynomial and pointwise properties established in Proposition 6.11.

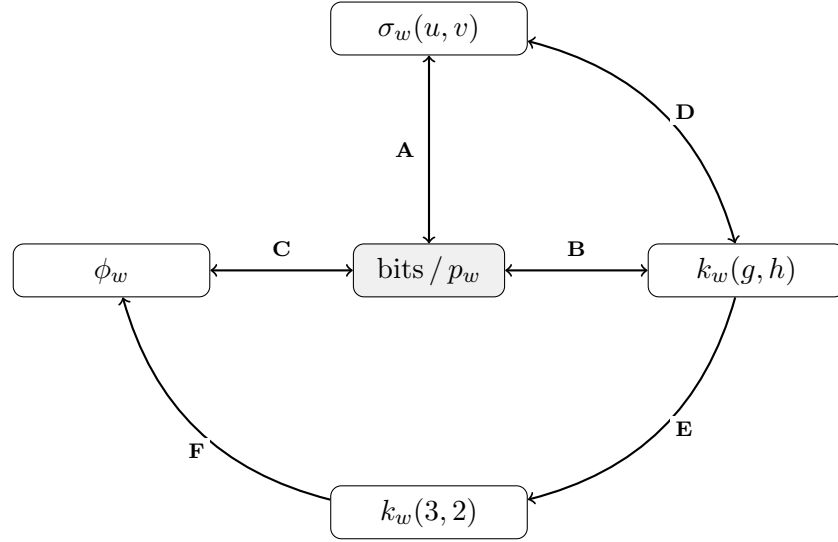
Definition 8.1 (Orbit parity sequence ϕ_w). Let w be a parity sequence with o odd steps, e even steps, and $k_w(3, 2) = k$. Set $q = 2^e - 3^o$ and define the sequence $a_0 = k$, $a_{i+1} = \Phi_{\text{step}}(a_i)$ under the $(3, 2, q)$ Collatz map. The *orbit parity sequence* of w is the parity sequence

$$\phi_w := (s_0, s_1, \dots, s_{n-1}), \quad s_i = \begin{cases} O & \text{if } a_i \text{ is odd,} \\ E & \text{if } a_i \text{ is even.} \end{cases}$$

A word on what *invertible* means here. Invertibility does not require that every edge admit a direct algebraic inverse. It requires only that every representation can be recovered from every other by composing forward transformations along some path in the diagram. Each forward transformation is lossless: no information about w is discarded, so any path that returns to the originating node achieves a genuine inverse. In particular, edge E (specialisation to $(3, 2)$) has no direct inverse, but the composition $F \circ C \circ B$ returns to $k_w(g, h)$ via

three lossless forward steps. The result is a true inverse of E, not an approximation. This is the ordinary transitivity of function composition: f is invertible whenever there exist functions $g_1, g_2, \dots$ such that $g_k \circ \dots \circ g_1 \circ f = \text{id}$, regardless of how many steps the composition requires. Since $k_w(3, 2)$ does not cover all of $\mathbb{N}$ — only those integers that arise as structural polynomials of valid parity sequences — edge E is a bijection onto its image. On that image, $F \circ C \circ B$ is both a left and right inverse of E, making it a two-sided inverse on the domain of valid parity sequence representations.

Every parity sequence w of length n with o odd steps lives simultaneously in these five representations, connected by explicitly invertible transformations. The canonical integer encoding (bits, p_w) sits at the centre; four structural representations — $\sigma_w(u, v)$, $k_w(g, h)$, $k_w(3, 2)$, and ϕ_w (Definition 8.1) — surround it. The diagram below serves as a navigational frame for the subsections that follow; each subsection develops one or more nodes and the edges connecting them.



Edge	Pair	Forward	Inverse
A	bits $\leftrightarrow$ σ_w	$\sigma_w(u, v) = \sum_{j=0}^{o-1} u^{o-1-j} v^{p_j}$	$p_w = \sigma_w(1, 2) + 2^n$
B	bits $\leftrightarrow$ $k_w(g, h)$	$k_w(g, h) = \sum_{j=0}^{o-1} g^{o-1-j} h^{p_j}$	$p_w = 2^{o-1} k_w(\frac{1}{2}, 2) + 2^n$
C	bits $\leftrightarrow$ ϕ_w	$\phi_w[i] = O$ if $b_i = 1$, else $\phi_w[i] = E$	$b_i = 1$ if $\phi_w[i] = O$, else $b_i = 0$
D	$\sigma_w \leftrightarrow k_w(g, h)$	$k_w(g, h) = \sigma_w(gh, h)/h^{o-1}$	$\sigma_w(u, v) = k_w(u/v, v) \cdot v^{o-1}$
E	$k_w(g, h) \rightarrow k_w(3, 2)$	$(g, h) \mapsto (3, 2)$	(via F then C then B)
F	$k_w(3, 2) \rightarrow \phi_w$ (Def. 8.1)	$q = 2^e - 3^o$, $a_0 = k_w(3, 2)$; iterate $(3, 2, q)$ map	(via C then B then E)

8.1 The p -value (edges B, C; centre node)

Since k_w is independent of trailing E -steps (Remark 2.8), it does not uniquely identify a parity sequence. A simple specialisation recovers a canonical integer encoding, and a sentinel bit resolves the trailing- E ambiguity.

Proposition 8.2 (Bit-encoding of parity sequences). *Let $w = b_0b_1 \cdots b_{n-1}$ be a parity sequence of length n , where $b_i = 1$ if the i -th step is an O -step and $b_i = 0$ if it is an E -step. Then*

$$N_w := 2^{o-1} \cdot k_w\left(\frac{1}{2}, 2\right) = \sum_{i=0}^{n-1} b_i 2^i,$$

the non-negative integer whose binary representation encodes w directly (LSB-first, $O = 1$, $E = 0$).

Proof. The j -th term of $k_w(g, h)$ is $g^{o-1-j} h^{\kappa_j}$. Evaluating at $g = 1/2$, $h = 2$ and multiplying by 2^{o-1} :

$$2^{o-1} \cdot \left(\frac{1}{2}\right)^{o-1-j} \cdot 2^{\kappa_j} = 2^{j+\kappa_j}.$$

Since κ_j counts E -steps before the j -th O -step and j counts O -steps before it, $j + \kappa_j = p_j$ is the position of the j -th O -step, i.e. the index i with $b_i = 1$. Summing over j gives $\sum_j 2^{p_j} = \sum_i b_i 2^i = N_w$. $\square$

Sequences differing only in trailing E -steps (i.e. trailing $b_i = 0$) share the same N_w . To resolve this ambiguity we append a sentinel bit at position n .

Definition 8.3 (p -value). The p -value of a parity sequence $w = b_0b_1 \cdots b_{n-1}$ is

$$p_w := \sum_{i=0}^{n-1} b_i 2^i + 2^n = 2^{o-1} \cdot k_w\left(\frac{1}{2}, 2\right) + 2^n.$$

Theorem 8.4 (p -value uniqueness). *The map $w \mapsto p_w$ is a bijection from the set of all finite parity sequences over $\{O, E\}$ (including the empty sequence) to the set of integers ≥ 1 . The empty sequence has $p_\varepsilon = 2^0 = 1$. For non-empty w , w is uniquely recovered from p_w by writing p_w in binary: the highest set bit (at position n) records the length, and bits $0, \dots, n-1$ give the sequence with $1 = O$ and $0 = E$.*

Proof. The empty sequence has $n = 0$, $N_\varepsilon = 0$, and $p_\varepsilon = 0 + 2^0 = 1$. For non-empty w , given $p_w \geq 2$, let $n = \lfloor \log_2 p_w \rfloor \geq 1$ (the position of the highest set bit). Then bits $0, \dots, n-1$ of p_w coincide with N_w and uniquely determine the positions of all O -steps, hence the full sequence w . The map is therefore injective. Surjectivity onto integers ≥ 1 follows since every positive integer has a unique highest set bit at some position $n \geq 0$: $n = 0$ gives the empty sequence, and $n \geq 1$ gives the parity sequence whose i -th step is O iff bit i of p_w is 1. $\square$

Remark 8.5 (Successor as cyclic bit rotation). The successor $w \mapsto w'$ (Proposition 6.12) acts on the p -value as a *right* rotation of the lower n bits. Reading the data bits in the conventional MSB-first order as $b_{n-1}b_{n-2} \cdots b_1b_0$, the right rotation moves the rightmost bit b_0 to the leftmost data position:

$$p_{w'} = 2^n + b_0 2^{n-1} + b_{n-1} 2^{n-2} + \cdots + b_1.$$

The sentinel bit at position n is unchanged, reflecting the fact that all cycle representatives have the same length n .

8.2 The σ -polynomial (edges A, D)

Change of variables

The σ -polynomial arises from $k_w(g, h)$ by the substitution $g = u/v$, $h = v$, followed by clearing the denominator v^{o-1} : $\sigma_w(u, v) = k_w(u/v, v) \cdot v^{o-1} \in \mathbb{Z}[u, v]$. The inverse substitution $u = gh$, $v = h$ recovers $k_w(g, h) = \sigma_w(gh, h)/h^{o-1}$.

Definition 8.6 (σ -polynomial). For a parity sequence w of length n with o odd steps and O -step positions $p_0 < \dots < p_{o-1}$, define

$$\sigma_w(u, v) := k_w\left(\frac{u}{v}, v\right) \cdot v^{o-1} = \sum_{j=0}^{o-1} u^{o-1-j} v^{p_j} \in \mathbb{Z}[u, v].$$

The exponent of u in the j -th term counts the O -steps *after* position j ; the exponent of v is the bit-position of the j -th O -step. Specialising to $u = 1$, $v = 2$ recovers the bit-encoding: $N_w = \sigma_w(1, 2)$, and consequently $p_w = \sigma_w(1, 2) + 2^n$ (edge A inverse).

The change of variables also induces a natural rotation operator on σ_w , corresponding to the k_w successor of Proposition 6.12 under the same substitution.

The successor operation on σ

Each step of the Collatz map acts on $\sigma_w(u, v)$ by an elegant polynomial operation [1]. Define

$$\gamma_w := \begin{cases} 1 & \text{if } p_0 = 0 \text{ (i.e. } w \text{ begins with an } O\text{-step),} \\ 0 & \text{otherwise.} \end{cases} \quad (\gamma)$$

Equivalently, $\gamma_w = 1$ if and only if the leading term $u^{o-1}v^{p_0}$ of σ_w has v -exponent zero. Define

$$\text{succ}_\sigma(\sigma_w(u, v)) := u^{\gamma_w} \cdot \sigma_w(u, v) \cdot v^{-1}.$$

Then, in the quotient ring $\mathbb{Z}[u, v]/(u^o - 1, v^n - 1)$,

$$\text{succ}_\sigma(\sigma_w(u, v)) = \sigma_{w'}(u, v),$$

where w' is the parity sequence obtained by rotating w left by one step (moving the first bit to the last position). The relation $v^n = 1$ is essential: the factor v^{-1} introduced at each step accumulates to $v^{-n} = 1$ after a full cycle of n steps, returning to σ_w . The relation $u^o = 1$ is essential for the same reason with respect to the O -step count. This identity does *not* hold as a free polynomial identity in $\mathbb{Z}[u, v]$.

Remark 8.7. Adopting the convention $0^0 = 1$, we have $\gamma_w = \sigma_w(1, 0)$: each term $1^{o-1-j} \cdot 0^{p_j}$ vanishes unless $p_j = 0$, which occurs precisely when $j = 0$ and w begins with an O -step. Complex roots of unity $u = \omega_o$, $v = \omega_n$ are one natural realisation of the quotient-ring constraints.

Evaluation at roots of unity

Proposition 8.8. *Let $\omega_o = e^{2\pi i/o}$ and $\omega_n = e^{2\pi i/n}$ be primitive o -th and n -th roots of unity respectively. The evaluation*

$$\sigma_w(\omega_o, \omega_n) = \sum_{j=0}^{o-1} \omega_o^{o-1-j} \omega_n^{p_j}$$

expresses the parity sequence as a linear combination of n -th roots of unity, weighted by powers of ω_o . Each step of the cycle acts as

$$\text{succ}_\sigma(\sigma_{w_k}(\omega_o, \omega_n)) = \omega_o^{\gamma_{w_k}} \cdot \sigma_{w_k}(\omega_o, \omega_n) \cdot \omega_n^{-1},$$

a multiplication by $\omega_o^{\gamma_{w_k}}$ (recording whether step k is an O -step) composed with a rotation ω_n^{-1} of the phase. Since $\omega_o^o = 1$ and $\omega_n^n = 1$, the cycle closes after exactly n steps. See [1] for the full development.

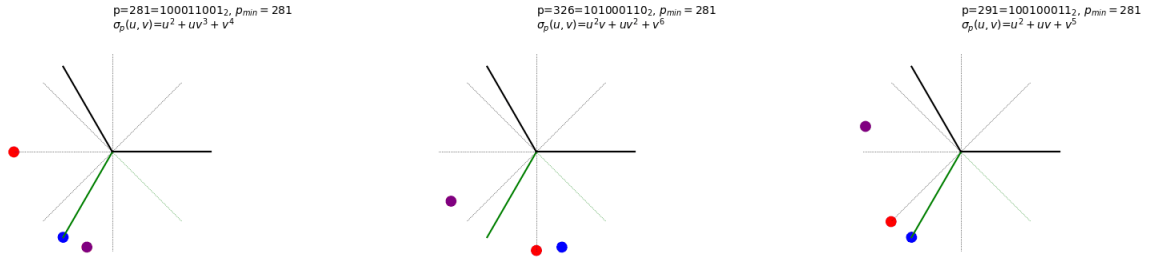


Figure 1: Three successive steps of the σ -successor for $w = OEEOOEEE$ ($n = 8, o = 3$), evaluated at (ω_3, ω_8) . Coloured dots show the three monomials of σ_w ; the green arrow is their sum (the value of σ_w as a complex number). *Left* (step 0, O -step, $\gamma = 1$): $\sigma = u^2 + uv^3 + v^4$. The O -step multiplies by ω_3 in addition to rotating by ω_8^{-1} , changing the u -weights as well as the phase. *Centre* (after two E -steps): $\sigma = u^2v + uv^2 + v^6$ — the dots have rotated twice; the sum (green arrow) has changed direction. *Right* (after the second O -step, $\gamma = 1$): $\sigma = u^2 + uv + v^5$. The O -step multiplies by ω_3 in addition to rotating by ω_8^{-1} , redistributing the u -weights and visibly changing the triangle of dots.

8.3 Worked example (all edges, $w = OEEOEOEE$)

The remaining nodes $k_w(3, 2)$ and ϕ_w are reached via edges E and F respectively; the return path from ϕ_w runs via **C** then **B** then **E**.

C⁻¹ $b_8 b_7 \cdots b_0 = 1, 0, 0, 1, 0, 1, 0, 0, 1$ ($O=1, E=0, b_8=1$ sentinel)
 O -positions $p_0=0, p_1=3, p_2=5$

A $\sigma_w(u, v) = u^2 + u v^3 + v^5$

A⁻¹ $p_w = \sigma_w(1, 2) + 2^8 = 41 + 256 = 297 = 100101001_2$

D $k_w(g, h) = \sigma_w(gh, h)/h^2 = g^2 + gh^2 + h^3$

B⁻¹ $p_w = 2^{o-1} k_w(\frac{1}{2}, 2) + 2^8 = 41 + 256 = 297 = 100101001_2 \checkmark$

E $k_w(3, 2) = 3^2 + 3 \cdot 2^2 + 2^3 = 29$

F $q = 5, a_0 = 29$; iterate $(3, 2, 5)$:

$$29 \xrightarrow{O} 92 \xrightarrow{E} 46 \xrightarrow{E} 23 \xrightarrow{O} 74 \xrightarrow{E} 37 \xrightarrow{O} 116 \xrightarrow{E} 58 \xrightarrow{E} 29 \Rightarrow \phi_w = OEEOEOEE$$

C $b_8 b_7 \cdots b_0 = 1, 0, 0, 1, 0, 1, 0, 0, 1$ ($b_8=1$ sentinel encodes $n=8$)

B $\kappa_j = (0, 2, 3), k_w(g, h) = g^2 + gh^2 + h^3 \checkmark$

Since $\gcd(5, 29) = 1$, the canonical basis gives $q_w = 5, a_w = 29$: no $3x+1$ cycle, but an integer $3x+5$ cycle at $a = 29$.

9 Examples

Example 9.1 (The $5x+1$ cycle at 17). Under the Syracuse $5x+1$ map (basis $g = 5, h = 2, q = 1$), the odd integer 17 is a periodic point. Its orbit is

$$17 \xrightarrow{O} 86 \xrightarrow{E} 43 \xrightarrow{O} 216 \xrightarrow{E^3} 27 \xrightarrow{O} 136 \xrightarrow{E^3} 17.$$

The clause is $w = OEEOEOEOEE$, with $o = 3, e = 7$.

The κ -tuple is $\kappa_0 = 0, \kappa_1 = 1, \kappa_2 = 4$ (the three O -steps are preceded by 0, 1, and 4 even steps respectively). Note that $(\kappa_0, \kappa_1, \kappa_2) = (0, 1, 4) \neq (0, 1, 2)$, so w is *not* a Steiner word and the geometric polynomial theorem does not apply directly. The structural polynomial is

$$k_w(g, h) = g^2 h^0 + g^1 h^1 + g^0 h^4 = g^2 + gh + h^4.$$

Evaluating at $(g, h) = (5, 2)$:

$$k_w(5, 2) = 25 + 10 + 16 = 51, \quad h^e - g^o = 2^7 - 5^3 = 128 - 125 = 3.$$

By Theorem 6.8, $d \cdot a = q \cdot k$ gives

$$3 \cdot a = 1 \cdot 51 = 51, \quad a = 17. \quad \checkmark$$

The near-miss $h^e - g^o = 3$ is what makes this cycle arithmetically possible: $2^7 - 5^3 = 3$ is a specific arithmetic accident with no analogue for the Collatz basis $(g, h) = (3, 2)$. This also illustrates Proposition 6.11: the polynomial $h^7 - g^3$ does not divide $g^2 + gh + h^4$ in $\mathbb{Z}[g, h]$, yet divisibility holds at $(5, 2)$ because of this accident.

Example 9.2 (The $g = h^2 - 1$ family). When divisibility *does* hold as a polynomial identity, it explains cycles across an entire family of bases at once. Consider $\text{St}(1, 1)^3$ (three copies of OEE), with $o = 3, e = 6$. By the composition rule:

$$\begin{aligned} k_{\text{St}(1,1)^2}(g, h) &= g + h^2, \\ k_{\text{St}(1,1)^3}(g, h) &= g^2 + gh^2 + h^4. \end{aligned}$$

On the locus $g = h^2 - 1$ one checks that $(g^2 + gh^2 + h^4) - (h^6 - g^3) = (g - h^2 + 1)(g^2 + gh^2 + h^4)$ vanishes identically on the locus $g = h^2 - 1$ (since $g - h^2 + 1 = 0$ there), so $k_{\text{St}(1,1)^3} = h^6 - g^3$ and the cycle element $a = q \cdot k / (h^6 - g^3) = q$ for every such basis. At $q = 1$:

- $h = 2, g = 3$: orbit $1 \xrightarrow{O} 4 \xrightarrow{EE} 1$, three times.
- $h = 3, g = 8$: orbit $1 \xrightarrow{O} 9 \xrightarrow{EE} 1$, three times.

Example 9.3 (Non-Collatz parity sequence). This example relaxes the requirement that the κ -tuple be strictly increasing. In the integer domain this corresponds to relaxing the rule that $gx + q$ is applied only to odd integers; in both settings the result is an “illegal” cycle that does not arise in standard Collatz dynamics.

Consider the parity sequence $w = OEEOOEEE$, with $o = 3, e = 5$. The κ -tuple is $\kappa_0 = 0, \kappa_1 = 2, \kappa_2 = 2$: both the second and third O -steps are preceded by exactly 2 even steps. The structural polynomial is

$$k_{OEEOOEEE}(g, h) = g^2 h^0 + g^1 h^2 + g^0 h^2 = g^2 + h^2(g + 1).$$

This w is *not* a valid standard Collatz parity sequence. The equal values $\kappa_1 = \kappa_2 = 2$ mean that two consecutive O -steps occur without any intervening E -step—the subsequence OO appears. Under standard Collatz, an O -step must be applied to an odd integer, so two consecutive O -steps are impossible: after $3n + 1$ the result is even and must be halved before another O -step can occur. Equivalently, valid Collatz sequences require $\kappa_0 < \kappa_1 < \dots < \kappa_{o-1}$ (strictly increasing).

If one relaxes this single condition—allowing $\kappa_j \leq \kappa_{j+1}$ —and correspondingly drives the $3x + 1$ and $x/2$ operations by the letters of w rather than the parity of the current element, then cycles can exist. At $(g, h) = (3, 2)$ we get $k_w(3, 2) = 3^2 + 2^2(3 + 1) = 9 + 16 = 25$ and $h^e - g^o = 2^5 - 3^3 = 5$, so Theorem 6.8 yields cycle element $a = 25/5 = 5$. The orbit from $a = 5$ is

$$5 \xrightarrow{O} 16 \xrightarrow{E} 8 \xrightarrow{E} 4 \xrightarrow{O} 13 \xrightarrow{O} 40 \xrightarrow{E} 20 \xrightarrow{E} 10 \xrightarrow{E} 5,$$

which does return to 5. The single illegal step is $4 \xrightarrow{O} 13$: applying $3x + 1$ to the even integer 4, forced by the repeated value $\kappa_1 = \kappa_2 = 2$.

Remark 9.4. The sequence $OEEOOEEE$ above has p -value $N_w + 2^8 = 25 + 256 = 281$: the integer 281 encodes the sequence directly in binary (with sentinel bit at position 8), and the glitched orbit was constructed by reading the bits of $281 - 256 = 25$ as a parity sequence.

A full end-to-end illustration of all five representations (edges A–F of the diagram in Section 8) for a single word appears in the worked example of Section 8. The examples above focus on the structural polynomial and cycle element identity across a range of bases and word types. Section 10 reflects on what this framework, taken as a whole, reveals — and what it cannot.

10 What the Abstract Path Tells Us — and What It Does Not

The preceding sections have established that a parity sequence w admits many equivalent representations: as the orbit parity sequence ϕ_w , as a binary integer p_w , as the σ -polynomial

$\sigma_w(u, v)$, as the structural polynomial $k_w(g, h)$ over an arbitrary basis, and as the integer $k_w(3, 2)$. When w is a cycle word, it carries additionally the cycle flavour table of Section 7. Each of these representations illuminates a different facet of the same object, as the examples of Section 9 illustrate concretely: the $5x + 1$ cycle at 17 (Example 9.1) and the $g = h^2 - 1$ family (Example 9.2) both arise as specialisations of a single abstract parity sequence to different bases.

Ironically, one of the least revealing representations of a cycle is the one most commonly encountered in the literature: a cycle expressed as a list of integers in a specific basis (g, h, q) . When $k = k_w(g, h)$ and $d = h^e - g^o$ are collapsed into a single ratio $a = qk/d$, most of the algebraic structure disappears. The cycle element a is just a number. The richness encoded in k_w as a polynomial in g and h — its monomial structure, the κ -tuple, the geometric sum form for Steiner words — is invisible. That the structure survives at all, and that a is an integer, is one of the small mysteries of mathematics; but the *why* is not apparent from the integer alone.

By contrast, the abstract representations make structure explicit. The κ -tuple encodes exactly how E-steps are distributed before each O-step. The structural polynomial $k_w(g, h)$ makes the dependence on the basis parameters transparent: one can see immediately how the cycle element scales as g or h varies, and how cycles at different bases relate to one another. The σ -polynomial connects the combinatorial structure of w to geometry: the monomials are points in a lattice, and the successor operation is a rotation. The p -value gives a canonical integer encoding that is independent of any basis.

All integer Collatz-like cycles are mere encodings of an abstract cycle

Once a parity sequence w and an encoding basis (g, h) are chosen, everything else is determined. The cycle element $a_w = k_w(g, h) / \gcd(k_w(g, h), |d|)$ and the additive parameter $q_w = d / \gcd(k_w(g, h), |d|)$ follow with no further freedom. There are no free integer cycles: for any parity sequence w and basis (g, h) , the parameters (a_w, q_w) are fixed by (w, g, h) through the reduced cycle element identity — there is no further freedom. Conversely, every reduced integer cycle (a, q) at a basis (g, h) arises from the parity sequence w that its orbit traces, so the map $(w, g, h) \mapsto (a_w, q_w)$ is surjective onto reduced integer cycles.

This perspective also reveals shared structure across different bases. Two bases (g_1, h_1, q_1) and (g_2, h_2, q_2) that yield the same parity sequence w as a cycle are encoding the same abstract object. The $g = h^2 - 1$ family (Example 9.2) makes this vivid: the bases $(g, h) = (3, 2)$ and $(g, h) = (8, 3)$ share the clause $\text{St}(1, 1)^3$ as a cycle word, differing only in the specialisation of the structural polynomial. More generally, the structural polynomial $k_w(g, h)$ makes this commonality explicit: any algebraic identity that holds in $\mathbb{Z}[g, h]$ — such as the factorisation $k_w(g, h) = h^e - g^o$ on the variety $g = h^2 - 1$ — is inherited by every basis that specialises to that variety.

The polynomial form can characterise individual bases

That said, the polynomial form can establish non-trivial facts that are *specific to the* $(3, 2, 1)$ *encoding*. A striking example is the $2a+1$ composition identity: for the Steiner words $\text{St}(\alpha, 1)$

and $\text{St}(1, \beta)$, the identity

$$\Phi_{\text{St}(1, \beta)}(\Phi_{\text{St}(\alpha, 1)}(a)) = \Phi_{\text{St}(\alpha+1, \beta+2)}(2a+1)$$

holds for all $\alpha \geq 1$, $\beta \geq 0$, and all a in the domain of $\Phi_{\text{St}(\alpha, 1)}$. In [4] we prove — working entirely in $\mathbb{Z}[g, h]$ via the structural polynomial — that among all bases (g, h, q) with $g, h \geq 2$ and $g \neq h$, this identity characterises $(g, h, q) = (3, 2, 1)$ uniquely. The proof forces $h = 2$ from the coefficient of a , then $g = 3$ by requiring the constant-term equation to hold for all α , and finally $q = 1$ by substitution. This is a purely polynomial argument: no numerical verification, no modular arithmetic, no appeal to orbit dynamics. The structural polynomial framework is the natural setting for such uniqueness results.

The abstract framework has limits, however. It does not, by itself, answer the question of whether the standard Collatz map $(3, 2, 1)$ has a non-trivial cycle. That question requires understanding the modular arithmetic of $3x+1$, $x/2$ — the residue structure, the admissibility constraints, the interplay between the parity of $3^o a + k_w(3, 2)$ and the required E-steps.

The limits are not total, however. The structural polynomial does carry modular information, and only a finite truncation is needed to extract it. Every monomial $g^{o-1-j} h^{\kappa_j}$ with $j \geq 3$ has $\kappa_j \geq j \geq 3$ (since the κ -tuple is non-decreasing and $\kappa_j \geq j$ for all j), so every such monomial is divisible by h^3 and contributes a multiple of h^3 to $k_w(g, h)$. The first three terms alone,

$$g^{o-1} + g^{o-2} h^{\kappa_1} + g^{o-3} h^{\kappa_2} \pmod{h^3},$$

determine $k_w(g, h) \pmod{h^3}$ completely — for any basis. At $(g, h) = (3, 2)$ this reads off the mod-8 character of the cycle element; at $(g, h) = (8, 3)$ the same truncation reads off the mod-27 character. The polynomial carries both facts simultaneously, as specialisations of a single identity in $\mathbb{Z}[g, h]$.

The abstract framework reveals the *form* of the modular constraint — the mod- h^3 truncation above is a concrete instance — but deciding whether that constraint is the right tool for attacking the cycle question at a specific basis requires engaging with the arithmetic of that basis. The mod-8 structure of $(3, 2, 1)$ and the mod-27 structure of $(8, 3, 1)$ are genuinely different, and the details matter.

In short: the abstract path framework is the right language for understanding *what* cycles are and *how* they relate to one another across bases. It is not sufficient on its own for understanding *whether* a particular cycle exists for a *specified* q : for any parity sequence w and any (g, h) , a cycle always exists if one is prepared to let q be determined by (w, g, h) via the reduced cycle element identity, but whether that q equals a pre-specified value — such as $q = 1$ for the standard Collatz map — is a question the abstract form alone cannot settle. Both levels of description are necessary, and they overlap more than the literature typically acknowledges.

11 Related Work

The structural polynomial $k_w(g, h)$ is closely related to the k -parameter appearing in the standard affine representation of Collatz orbits, where it appears as an entry in the affine

transformation matrix. The Steiner word framework was developed in [2] as a natural decomposition of Syracuse orbits; the present paper provides its polynomial algebraic foundation.

The σ -polynomial $\sigma_w(u, v)$ and its successor operation at complex roots of unity were introduced in [1]; the present paper situates them within the broader five-representation framework and makes their connection to $k_w(g, h)$ via the change of variables $g = u/v$, $h = v$ explicit.

The composition rule (Theorem 4.1) and cycle element identity (Theorem 6.8) are applied in the companion paper [4] to prove a uniqueness theorem for the $2a + 1$ Steiner composition identity and to characterise its domain constraint in terms of 2-adic valuations. Structural constraints on Collatz cycles derived from Steiner sentence analysis appear in [3].

References

- [1] J. Seymour, *Exploring Cyclic Structures in Polynomial Rings with Applications to the Collatz Conjecture*, preprint, 2025. <https://wildducktheories.github.io/collatz/papers/14-uv-polynomial/uv-polynomial.pdf>
- [2] J. Seymour, *Steiner Word Decomposition and Branch Length Distribution*, preprint, 2026. <https://wildducktheories.github.io/collatz/papers/67-branch-length-distribution/67-branch-length-distribution.pdf>
- [3] J. Seymour, *Structural Constraints on Collatz Cycles via Steiner Sentence Analysis*, unpublished, 2026.
- [4] J. Seymour, *The $2a + 1$ Composition Identity: Uniqueness and Domain Constraint*, unpublished, 2026.